

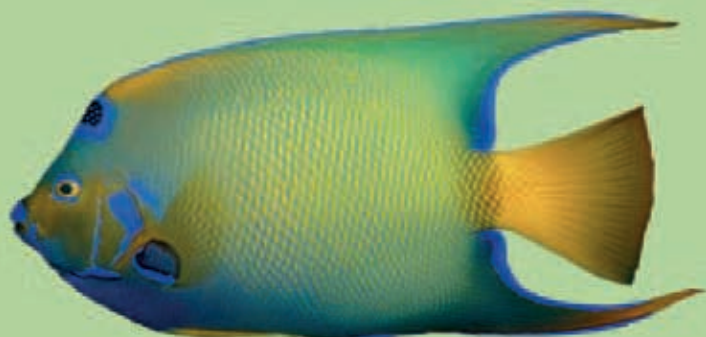
Scales

Many different animals have a flexible body covering of scaly skin. Scales are small, tough plates that grow from the skin in neat patterns. They give body protection and also help some animals to move. Insect scales are made of chitin, like the rest of their exoskeletons. The scales of vertebrates are made of hard substances such as dentine, which is also found in teeth, and the protein keratin, which also makes up claws and fingernails. Animals with the strongest scales include crocodiles, whose scales are reinforced with bone.



Cutting edge

It may look like a walking pine cone, but this scaly animal is actually a tropical mammal called a pangolin. Its scales are thin but hard-edged, and the pangolin curls up and raises them if it is attacked. Newborn pangolins have soft scales, but these start to harden after a few days. Once a pangolin is fully grown, its scales are kept sharp by wear and tear as it moves around in the forest trees.



Angelfish

Scaly jacket

Scales surround a fish's body like a tight-fitting jacket, helping it to slip easily through the ocean. Fish living in the open sea often have scaly skin with a silvery sheen, whereas species from coastal waters and coral reefs are usually brilliantly colored. Growth rings on the scales of bony fish can be used to estimate a fish's age.



Shedding skin

Snake skin has a thin outer layer, or cuticle, which is very smooth to the touch. The cuticle does not stretch and must be cast off so that the animal can grow. Snakes shed their whole cuticle in one piece, pulling it inside out by rubbing their bodies against rough objects.

Sandpaper skin

If a shark's skin is rubbed the wrong way, it feels as rough as sandpaper—at one time, people used it for exactly this purpose. Unlike other fish, a shark's body is covered in tiny toothlike scales. Known as denticles, they reduce turbulence in the water, letting sharks close in silently on their prey.



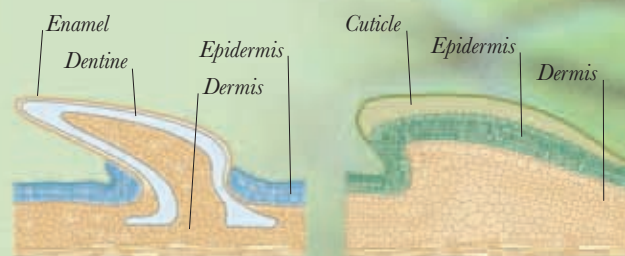
Microscopic view of shark's denticles



Great white shark

Scale structure

All scales are fixed in skin, but they are made of different materials and grow in different ways.



Shark scale

A shark's scales are made of dentine and enamel—hard tissues found in teeth—and are anchored in the deepest layer of the skin, or dermis. Each scale has a working life of a few weeks. After this, it is replaced.

Snake scale

The scales of snakes and lizards grow on the epidermis—the outer layer of the skin. They can lie flat or overlap in rows. These scales last for life, although their outer cuticle is shed several times a year.